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Deep Transfer Learning-Based Heart Failure Drug Classification on Small Datasets

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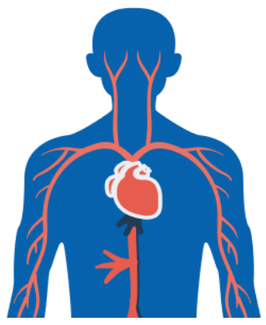
ABSTRACT

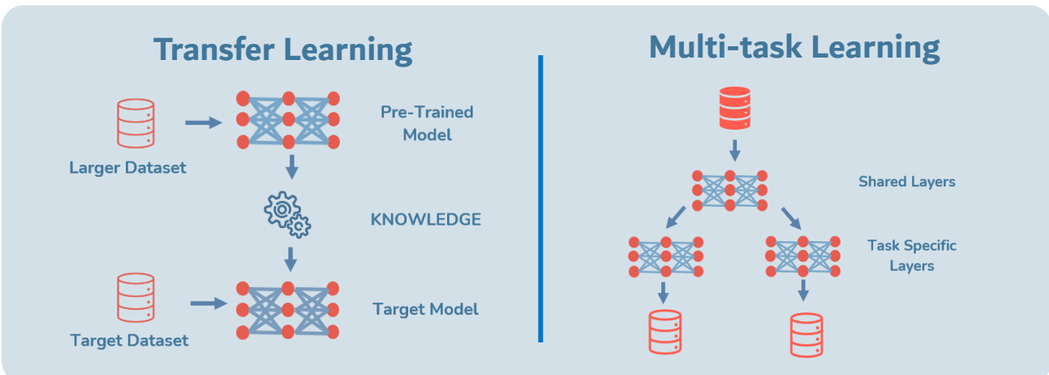
Despite heart failure's global significance, medications are still ineffective and low as drug discovery processes are long, costly, and especially complicated due to the pandemic. To overcome these challenges, a machine learning model which classifies active compounds that can inhibit the activities of disease-related receptors based on their pX bioactivity values using deep transfer learning techniques has been constructed. The pre-trained model was constructed. Compound datasets were labelled as active or inactive. SMILES were converted into molecular fingerprints before undergoing FP2VEC feature extraction process. Afterwards, the vectors were used for model training hyperparameters embedding size (k), window size of filter (h) and size of feature map (n) were hypertuned. The model at k = 50, h = 5 and n = 1024 when trained with datasets formed by using an N = 1000 nM bioactivity value. The model performed average AUC scores of 0.858, a higher score compared to the single task model. Afterwards, weight and bias will be transferred before target model training. Once completed, the model will hopefully assist heart failure drug discoveries, benefit patients and the pharmaceutical industry.

1. INTRODUCTION



Heart failure is one of the main contributing causes of death worldwide. The disease has led to over 26 million patients in each year and resulted in a huge economic burden of up to 108 billion US dollars. However, current medications are still ineffective and limited in numbers as they are going through long and costly drug discovery processes. Also, there are still only few computational researches and heart failure drugs datasets, making it difficult for machine learning model training.

2. THEORY



pX represents a compound's bioactivity value, or **how well a compound can prevent a receptor's activity.** To overcome study challenges, two techniques were brought up; **transfer learning and multi-task learning**.

3. METHODOLOGY

1. Data Preparation

Compound datasets of three protein families related to heart failure were downloaded from Reaxys database where irrelevant information is then removed. Each compound was labelled with a type according to its pX bioactivity value (pX = -log X) as labelled in the downloaded dataset. The criteria pX used for type classification was varied with N = 100 nM, 300 nM, 500 nM and 1000 nM through the method as shown below in Figure 1.

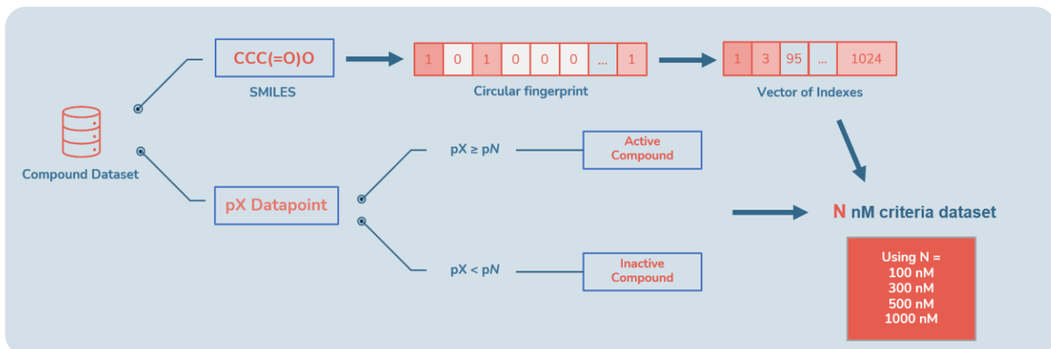


Figure 1. Dataset Formation Process

2. Feature Extraction

For feature extraction, each compound's SMILES data was converted to a 1024 circular molecular fingerprint. The feature of each fingerprint was extracted using the FP2VEC technique.

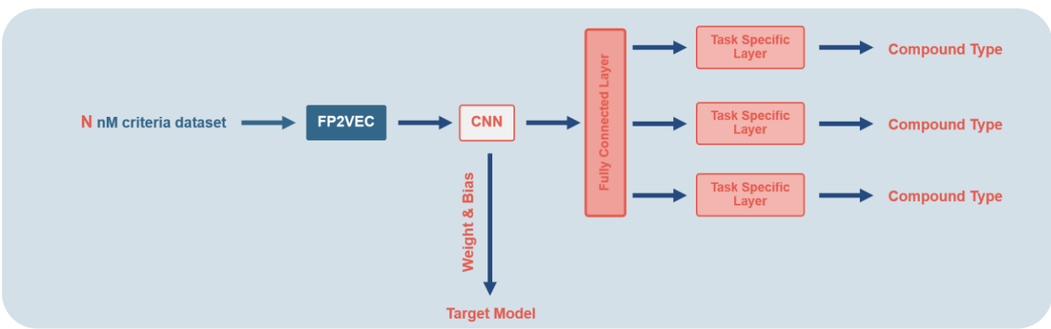


Figure 2. Pre-Trained Model Architecture

3. Model Training

The pre-trained model, with architecture as shown in Figure 2, is trained using the gained vector of indexes in the form of a multi-task model. Hypertuning was performed on hyperparameters embedding size (k), window size of filter (h) and size of feature map (n). Hyper-parameters tuning were split into 3 rounds, by first varying the hyper-parameter k, h and then n, where the hyper-parameter that yielded the best results from the previous step was fixed during the later on other tuning rounds.

4. Performance Evaluation

The model's performance was evaluated using the area under curve or AUC of ROC curve (receiver operating characteristic curve). The AUC scores lay between 0 (indicating worst performance) to 1 (indicating best performance). The logarithm loss of each task was also calculated to observe the number of inaccurate predictions.

4. RESULT & DISCUSSION

N (nM)	Target Protein Family	Multi-Task Model		Single Task Model	
		AUC Score	Average AUC Score	AUC Score	Average AUC Score
1000	GPCR Subfamily A3	0.940	0.858	0.923	0.849
	Nuclear Receptor Subfamily 3	0.853		0.824	
	GPCR Subfamily A17	0.780		0.802	
500	GPCR Subfamily A3	0.899	0.838	0.878	0.819
	Nuclear Receptor Subfamily 3	0.859		0.829	
	GPCR Subfamily A17	0.756		0.751	
300	GPCR Subfamily A3	0.890	0.825	0.846	0.815
	Nuclear Receptor Subfamily 3	0.840		0.832	
	GPCR Subfamily A17	0.745		0.767	
100	GPCR Subfamily A3	0.911	0.847	0.884	0.818
	Nuclear Receptor Subfamily 3	0.859		0.803	
	GPCR Subfamily A17	0.771		0.767	

Table 1. Multi-Task & Single-Task Model AUC Scores at k = 50, h = 5, and n = 1024

The set of hyperparameters that yielded the model with the highest AUC scores was determined as k = 50, h = 5, and n = 1024 when using N = 1000 nM as a dataset classification criteria. Compared to the single task model version, our pre-trained/Multi-task model yield higher average AUC score. For the models' loss comparison, both models overfit while the multi-task model yield higher losses. As for comparison of both models, a higher loss suggests that the multi-task model's false predictions differ a lot from the actual outputs. More experiments will be conducted to observe this trend.

5. CONCLUSION & PLAN

Our multi-task model performed better than simple single task models for the problem. For future plans, target model training will begin after bias transfer. Further architecture improvements may be done to improve accuracy. After completed, the model will hopefully be able to assist drug discovery processes by reducing cost, time and difficulties in lab practices during the pandemic.

6. REFERENCE

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FULL RESULT

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